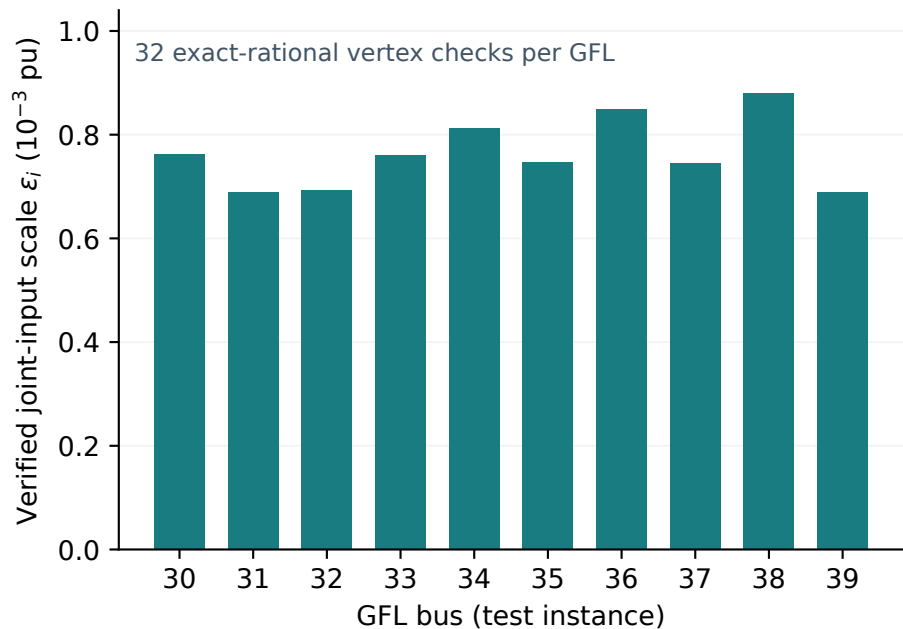


Local nonlinear guarantees need a network compatibility condition

Full GFL model: PLL + current loop + DC dynamics; filtered RoCoF instrument.

A Ten conditional invariant regions



An exact algebraic slice of the nonlinear DAE violates the assumed input envelopes of the product region.

Scope: a failure of this certificate composition. Instability and reachability from nominal operation are not established.

B Explicit network closure counterexample

